

CHAPTER 7

WELL ORDERED SETS

7.1. A well ordered set¹⁷ is a poset

$$U = (\text{Field}(U), \leq_U),$$

where $\leq_U$ is a wellordering on $\text{Field}(U)$, i.e., a linear (total) ordering on $\text{Field}(U)$ such that every non-empty $X \subseteq \text{Field}(U)$ has a least member. Associated with U is its **strict ordering** $<_U$,

$$x <_U y \iff_{\text{df}} x \leq_U y \ \& \ x \neq y.$$

For example, the set $\mathbb{N}$ of natural numbers is well ordered by its natural ordering, and so is each of its finite initial segments

$$[0, n) = \{i \in \mathbb{N} \mid i < n\}.$$

A “longer” well ordered set is $(\mathbb{N} \cup \{\infty\}, \leq')$, where ∞ is some object not in $\mathbb{N}$ which we put after all the natural numbers, i.e.,

$$x \leq' y \iff y = \infty \vee [x, y \in \mathbb{N} \ \& \ x \leq y].$$

As we did with arbitrary posets in the preceding chapter, we will usually identify U with its field, talk about *the points* or *subsets* of U , meaning the members and subsets of $\text{Field}(U)$, skip the subscript in $\leq_U$ or $<_U$ when it is obvious from the context, etc.

The most basic results of the last two chapters were all proved by some combination of the coupling

$$\text{definition by recursion} - \text{proof by induction.} \quad (7-1)$$

In the simplest case, some function $f : \mathbb{N} \rightarrow E$ is defined by recursion, some properties of f are proved by induction and these in turn imply the theorem we want. Typical are the Continuous Least Fixed Point and the Schröder-Bernstein theorems which say nothing (explicitly) about recursion, induction or any functions with domain $\mathbb{N}$, but whose proofs most assuredly use precisely these notions. We based the proof of the Recursion Theorem 5.6 directly on the Induction Axiom for the natural numbers. The key fact, however, which

¹⁷Do we dare call them *wosets*? It's not much worse than *posets* and it would sure save a lot of key strokes.



FIGURE 7.1. The low end of a long wellordering.

can be generalized is that $\mathbb{N}$ is well ordered by its natural ordering. Here we will generalize 5.6 to a powerful TRANSFINITE RECURSION THEOREM 7.24 which justifies *definition by recursion* of functions $f : U \rightarrow E$ on any well ordered set U . Coupled with HARTOGS' THEOREM 7.34 which guarantees the existence of "arbitrarily large" well ordered sets, this makes it possible to apply the basic idea of (7-1) in situations far removed from the natural numbers. Typical applications are the FIXED POINT THEOREM 7.35 and its corollary, the LEAST FIXED POINT THEOREM 7.36, which is just 6.21 without the countable continuity hypothesis.

7.2. A set A is **well orderable** if it admits a wellordering, so it is the field of some well ordered set $(A, \leq)$. One of the chief lessons of this chapter is that well orderable sets behave much better than arbitrary sets: for example any two of them are comparable in cardinality, either $A \leq_c B$ or $B \leq_c A$. In fact, *every set is well orderable*. Zermelo showed this in 1904, settling with one brilliant stroke the problem of Cardinal Comparability and a whole slew of related, regularity questions about arbitrary sets. We will prove Zermelo's Wellordering Theorem in the next chapter, after we introduce the Axiom of Choice on which it is based. It is worth pointing out here, however, that the mathematical content of this fundamental result is just the sum of the Transfinite Recursion and Hartogs' Theorems: the Axiom of Choice simply allows us to put the two together.

7.3. Exercise. If C is well orderable and $A \leq_c C$, then A is well orderable.

7.4. Exercise. If C is well orderable and there exists a surjection $f : C \twoheadrightarrow A$, then $A \leq_c C$, and hence A is also well orderable.

7.5. Successor and limit points. Every well ordered set U looks at its low end like an initial segment of $\mathbb{N}$. If it is not empty, it must have a least member which is typically denoted by 0 rather than $\perp$,

$$0 = 0_U =_{\text{df}} \text{the least element of } U. \quad (7-2)$$

It is pictured by a hollow square, the first point in Figure 7.1. Each $x \in U$ other than the maximum (which may or may not exist) has an element following it immediately,

$$S(x) = S_U(x) =_{\text{df}} \min_U \{y \in U \mid x < y\}. \quad (7-3)$$

The values of the partial function $S : U \rightarrow U$ are the **successor points** of U . In addition, U may have **limit points** which are above 0 but not the successor of anything:

$$\text{Limit}_U(x) \iff_{\text{df}} 0 < x \ \& \ (\forall u < x)(\exists v)[u < v < x]. \quad (7-4)$$

These are pictured by black boxes in Figure 7.1. The first limit point of U is typically denoted by

$$\omega = \omega_U =_{\text{df}} \min\{x \in U \mid \text{Limit}(x)\}, \quad (7-5)$$

when it exists, the points below it are the **finite points** and the points above it (including ω) are the **infinite points** of U . If U is infinite, then the function $\pi : \mathbb{N} \rightarrow U$ defined by the recursion

$$\begin{aligned} \pi(0) &= 0_U = \text{the least member of } U, \\ \pi(n+1) &= S_U(\pi(n)), \end{aligned} \quad (7-6)$$

is an order-preserving correspondence of $\mathbb{N}$ with the finite points of U .

7.6. Exercise. For each subset $I \subseteq U$ of a well ordered set U , the restriction

$$x \leq_I y \iff_{\text{df}} x \leq_U y \ \& \ x, y \in I$$

of $\leq_U$ to I is a wellordering, so that I is a well ordered set in its own right with this ordering.

7.7. Definition. A well ordered set U is an **initial segment** of V if $\text{Field}(U)$ is a downward closed subset of $\text{Field}(V)$ and $\leq_U$ is the restriction of $\leq_V$ to $\text{Field}(U)$:

$$\begin{aligned} U \subseteq V &\iff_{\text{df}} \text{Field}(U) \subseteq \text{Field}(V) \\ &\& (\forall y \in \text{Field}(U))(\forall x \leq_V y)[x \in \text{Field}(U)] \\ &\& (\forall x, y \in \text{Field}(U))[x \leq_U y \iff x \leq_V y]. \end{aligned} \quad (7-7)$$

Clearly V is an initial segment of itself, the **trivial** one. With each $y \in V$ we associate the **proper initial segment** of points strictly below y ,

$$\mathbf{seg}(y) = \mathbf{seg}_V(y) =_{\text{df}} \{x \in V \mid x <_V y\} \subsetneq V. \quad (7-8)$$

More precisely, this is the field of $\mathbf{seg}(y)$, but the ordering is determined by V and we will talk about initial segments as if they were just sets, as usual.

7.8. Exercise. If 0 is the least element of U , then $\mathbf{seg}(0) = \emptyset$, and if $x \in U$ has a successor, then

$$\mathbf{seg}(S(x)) = \mathbf{seg}(x) \cup \{x\}.$$

7.9. Proposition. A set I is an initial segment of a well ordered set U if and only if $I = U$ or for some $x \in U$, $I = \mathbf{seg}(x)$.

PROOF. If $I \subsetneq U$, let $x = \min(U \setminus I)$ so that immediately,

$$y \in \mathbf{seg}(x) \implies y < x \implies y \in I,$$

and to verify that $I = \mathbf{seg}(x)$, it is enough to prove

$$y \in I \implies y < x.$$

Towards a contradiction, if $y \in I$ but $y \not< x$, then we must have $x \leq y$, which implies $x \in I$ because I is downwards closed, contradicting the choice of x . The converse is trivial. $\dashv$

7.10. Exercise. *The family of initial segments of a well ordered set U is well ordered by the relation $\sqsubseteq$.*

The general idea is to view a well ordered set U as a generalization of the natural number sequence $0, 1, 2, \dots$, possibly shorter than or of equal length to $\mathbb{N}$, typically much longer. The particular members of U will be of little consequence: it is the *length* of the sequence in which we will be interested. We introduce here the general notion of isomorphism which relates posets with the same *shape*, the shape of a well ordered set being just a “length”.

7.11. Definition. A function $\pi : P \rightarrow Q$ from one poset into another is **order-preserving** if for all $x, y \in P$,

$$x \leq_P y \iff \pi(x) \leq_Q \pi(y);$$

a **similarity** is an order-preserving bijection $\pi : P \xrightarrow{\sim} Q$, and if one such exists we call P and Q **similar**, **order isomorphic** or **copies** of each other. We write

$$P =_o Q \iff_{\text{df}} (\exists \pi : P \xrightarrow{\sim} Q)[\pi \text{ is a similarity}].$$

The subscript o in $=_o$ stands for “order type”, a fancier expression for “shape”. Notice that by our general convention of talking about a poset as if it were its field, we write $\pi : P \xrightarrow{\sim} Q$ for similarities instead of the more explicit $\pi : \text{Field}(P) \xrightarrow{\sim} \text{Field}(Q)$.

7.12. Exercise. *Every order-preserving $\pi : P \rightarrow Q$ from one poset to another is monotone; but there exist monotone mappings which are not order-preserving.*

7.13. Exercise. *If P and Q are linear posets, then a function $f : P \rightarrow Q$ is order-preserving if and only if it is **strictly monotone**, i.e.,*

$$x <_P y \implies f(x) <_Q f(y).$$

In particular, order-preserving functions on well ordered sets are strictly monotone, and hence one-to-one.

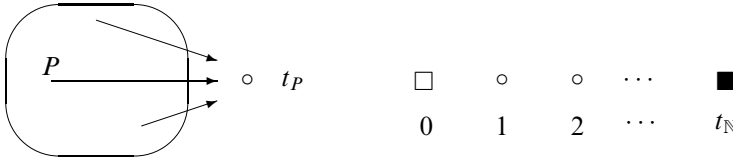
7.14. Exercise. *For all posets P, Q, R ,*

$$\begin{aligned} P &=_o P, \\ P &=_o Q \implies Q =_o P, \\ P &=_o Q \ \& \ Q =_o R \implies P =_o R. \end{aligned}$$

7.15. Lemma. *If a poset P is similar to a well ordered set U , then it is also well ordered.*

PROOF. Given $\emptyset \neq X \subseteq P$, let $p \in U$ be the $\leq_U$ -least element of the image $\pi[X]$ and verify (easily) that $x = \pi^{-1}(p)$ is $\leq_P$ -least in X , because π preserves the orderings. $\dashv$

We can construct explicitly some fairly long wellorderings by starting with $\mathbb{N}$ and its finite initial segments and applying repeatedly several natural operations on posets which yield well ordered sets on well ordered arguments. Here we look at just one of these, leaving the rest for the problems.

FIGURE 7.2. The successor poset to P and $\text{Succ}(\mathbb{N})$.

7.16. The **successor** $\text{Succ}(P)$ of a poset P is obtained by adding a new point above all the members of P . To be specific, we can choose to add to the field of P the object

$$t_P =_{\text{df}} r(\text{Field}(P)) \quad (7-9)$$

which is guaranteed by 3.11 to be a new element, and we set

$$x \leq_{\text{Succ}(P)} y \iff_{\text{df}} x \leq_P y \vee [x \in P \ \& \ y = t_P] \vee x = y = t_P. \quad (7-10)$$

If P is finite with n elements, then $\text{Succ}(P)$ has $n + 1$ elements, in fact, easily $\text{Succ}([0, n)) =_o [0, (n + 1))$. On the other hand, $\text{Succ}(\mathbb{N})$ is countably infinite, but with a different, “longer” ordering than $\mathbb{N}$, as it has a maximum element which comes after all the natural numbers.

7.17. Exercise. If $P =_o Q$, then $\text{Succ}(P) =_o \text{Succ}(Q)$.

7.18. Exercise. If U is well ordered, so is $\text{Succ}(U)$.

Using this successor operation on posets, we can view each well ordered set U as a proper initial segment of another,

$$U = \mathbf{seg}_{\text{Succ}(U)}(t_U) \subsetneq \text{Succ}(U). \quad (7-11)$$

7.19. Definition. A mapping $\pi : P \rightarrow P$ on a poset to itself is **expansive**, if for all $x \in P$, $x \leq \pi(x)$.

7.20. Theorem. Every order-preserving injection $\pi : U \rightarrow U$ of a well ordered set into itself is expansive.

PROOF. Towards a contradiction, assume that $\pi : U \rightarrow U$ is order-preserving but that for some $x \in U$, $\pi(x) < x$, and let

$$x^* = \min\{x \in U \mid \pi(x) < x\}.$$

Thus, $\pi(x^*) < x^*$, and so $\pi(\pi(x^*)) < \pi(x^*)$ since π is an order-preserving injection, which contradicts the choice of x^* . $\dashv$

7.21. Corollary. No well ordered set is similar with one of its proper initial segments, and hence no two distinct initial segments of a well ordered set are similar.

PROOF. Every similarity $\pi : U \rightarrow U$ is (in particular) an order-preserving injection of U into U , so we cannot have $\pi(x) < x$, by the theorem. $\dashv$

Because a well ordered set may have limit points in addition to its 0 and its successor points, it is easiest to generalize the principles of proof by *complete induction* and definition by *complete recursion*.

7.22. Transfinite Induction Theorem. *For every well ordered set U and every unary definite condition P ,*

$$\text{if } (\forall y \in U)[(\forall x < y)P(x) \implies P(y)] \text{ then } (\forall y \in U)P(y).$$

PROOF. Assuming the opposite, towards a contradiction, let

$$y^* =_{\text{df}} \min\{y \in U \mid (\forall x < y)P(x) \& \neg P(y)\};$$

the hypothesis yields $P(y^*)$, which contradicts the choice of y^* . $\dashv$

In specific cases, it is often just as easy to prove $(\forall y \in U)P(y)$ by contradiction rather than appeal to 7.22, in effect repeating this little argument. It depends on the statement to be proved and how much one is annoyed by dealing with negative statements. We will illustrate both styles. Incidentally, the term “transfinite” is used because U may be longer than $\mathbb{N}$, but the theorem also holds, of course, when U is finite or similar with $\mathbb{N}$.

The next lemma is the key step in the proof of the fundamental theorem which follows it.

7.23. Lemma. *Suppose U is a well ordered set and $h : (U \rightarrow E) \times U \rightarrow E$ maps the partial functions from U to E and U into E . It follows that for every $t \in U$, there exists exactly one function*

$$\sigma_t : \mathbf{seg}(t) \rightarrow E$$

which satisfies the identity

$$\sigma_t(x) = h(\sigma_t \upharpoonright \mathbf{seg}(x), x) \quad (x < t). \quad (7-12)$$

PROOF. By Transfinite Induction, assume that for each $u < t$ there exists exactly one function $\sigma_u : \mathbf{seg}(u) \rightarrow E$ such that

$$\sigma_u(x) = h(\sigma_u \upharpoonright \mathbf{seg}(x), x) \quad (x < u). \quad (7-13)$$

The induction hypothesis gives us nothing if $t = 0_U$ is the least point in U , but the required conclusion is trivial in this case taking $\sigma_0 = \emptyset$. If $t = Sv$ is a successor point in U , we set

$$\sigma_t = \sigma_v \cup \{(v, h(\sigma_v, v))\};$$

now (7-13) holds for $x < v$ by the induction hypothesis and it holds for $x = v$ by the definition (since $\sigma_t \upharpoonright \mathbf{seg}(t) = \sigma_v$). For the last case, when t is a limit point, we need a

Sublemma. *The set of functions $\{\sigma_u \mid u < t\}$ is a chain under $\subseteq$, i.e.,*

$$x < u < v < t \implies \sigma_u(x) = \sigma_v(x). \quad (7-14)$$

Proof. Assume not and let x be least such that (7-14) fails for some $u > x$, $u < v < t$. This means that

$$\sigma_u \upharpoonright \mathbf{seg}(x) = \sigma_v \upharpoonright \mathbf{seg}(x),$$

and then by the identity which σ_u, σ_v satisfy,

$$\begin{aligned}\sigma_u(x) &= h(\sigma_u \upharpoonright \mathbf{seg}(x), x) \\ &= h(\sigma_v \upharpoonright \mathbf{seg}(x), x) \\ &= \sigma_v(x),\end{aligned}$$

which contradicts the choice of x .

⊥ (Sublemma)

We now take

$$\sigma_t = \bigcup \{ \sigma_u \mid u < t \};$$

this is a function with domain $\mathbf{seg}(t)$ by the Sublemma, and it satisfies (7-12), since for each $x < t$,

$$\begin{aligned}\sigma_t(x) &= \sigma_u(x) && \text{for some } u \text{ such that } x < u < t, \\ &= h(\sigma_u \upharpoonright \mathbf{seg}(x), x) && \text{by ind. hyp., since } u < t, \\ &= h(\sigma_t \upharpoonright \mathbf{seg}(x), x) && \text{since } \sigma_u \upharpoonright \mathbf{seg}(x) = \sigma_t \upharpoonright \mathbf{seg}(x) \\ &&& \text{by the definition of } \sigma_t.\end{aligned}$$

This completes the proof of existence of σ_t , and its uniqueness is easily verified by Transfinite Induction, using (7-12). ⊥

7.24. Transfinite Recursion Theorem. *For each well ordered set U and each function $h : (U \rightarrow E) \times U \rightarrow E$, there exists exactly one function $f : U \rightarrow E$ which satisfies the identity*

$$f(x) = h(f \upharpoonright \mathbf{seg}(x), x) \quad (x \in U). \quad (7-15)$$

PROOF. Consider the well ordered set $\text{Succ}(U)$ which has some point $t = t_U$ on top of U , and the extension $h' : (\text{Succ}(U) \rightarrow E) \times \text{Succ}(U) \rightarrow E$ of h defined by

$$h'(\sigma, x) = \begin{cases} h(\sigma \upharpoonright U, x), & \text{if } x \in U, \\ e^*, & \text{otherwise, i.e., if } x = t, \end{cases}$$

where e^* is some arbitrary member of E , of no consequence. The function h' has the correct domain for applying the Lemma to $\text{Succ}(U)$ and h' , because $\mathbf{seg}_{\text{Succ}(U)}(t) = U$. For the top point t , the Lemma gives a unique function $f = \sigma_t : U \rightarrow E$ which satisfies (7-12) for all $x \in U$. ⊥

Perhaps the simplest, non-trivial application of Transfinite Recursion is the definition of transfinite orbits for mappings of an inductive poset into itself. We consider first the basic case of *expansive* mappings, defined in 7.19. Expansive mappings are related to the monotone mappings we studied in the last chapter, but the two notions do not coincide; for example, the constant mapping $X \mapsto \emptyset$ on $\mathcal{P}(\mathbb{N})$ is obviously monotone but not expansive, while

$$\pi(X) = \begin{cases} X \cup \{1\} & \text{if } 0 \in X, \\ X \cup \{2\} & \text{if } 0 \notin X \end{cases}$$

is expansive but (easily) not monotone. It turns out, however, that results about expansive mappings can often be translated into similar results about monotone mappings.

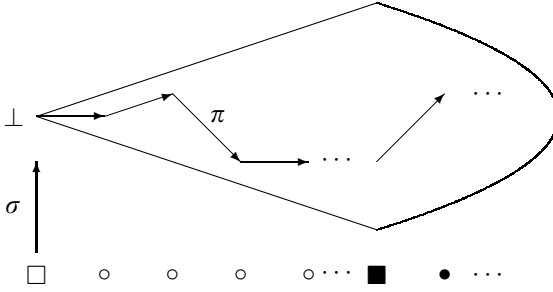


FIGURE 7.3. The transfinite orbit of an expansive mapping.

7.25. Iteration Lemma. Suppose $\pi : P \rightarrow P$ is an expansive mapping on an inductive poset and U is a well ordered set. There exists a unique function $\sigma : U \rightarrow P$ which satisfies the following conditions:

$$\begin{aligned} \sigma(0) &= \perp, \\ \text{if } x = S(y), \text{ then } \sigma(x) &= \pi(\sigma(y)), \\ \text{if } \text{Limit}(x), \text{ then } \sigma(x) &= \sup_P \{\sigma(y) \mid y < x\}. \end{aligned} \quad (7-16)$$

In addition, this σ is monotone from U to P , i.e.,

$$x \leq y \implies \sigma(x) \leq_P \sigma(y). \quad (7-17)$$

PROOF. The conditions in (7-16) just about give a definition of σ by transfinite recursion, except that there is a problem in the limit case if the set $\{\sigma(y) \mid y < x\}$ is not a chain in P . To account for this possibility, we define σ by appealing to Theorem 7.24 so that it satisfies the following:

$$\sigma(x) = \begin{cases} \perp, & \text{if } x = 0, \\ \pi(\sigma(y)), & \text{if } x = S(y) \text{ for some } y, \\ \sup_P \{\sigma(y) \mid y < x\}, & \text{if } \text{Limit}(x) \\ & \& (\forall x_1 < x_2 < x)[\sigma(x_1) \leq_P \sigma(x_2)], \\ \perp, & \text{otherwise,} \end{cases}$$

where $\leq = \leq_U$ is the wellordering of U , as in the statement of the theorem.

Sublemma. For each $x \in U$,

$$x_1 < x_2 \leq x \implies \sigma(x_1) \leq_P \sigma(x_2). \quad (7-18)$$

Proof. Assume not and let x be least in U such that (7-18) fails. Since (7-18) holds vacuously when $x = 0$ is the least element in U , we need consider only two cases.

CASE 1. $x = S(y)$ is a successor point. Assume $x_1 < x_2 \leq x$. If $x_2 \leq y$, we get $\sigma(x_1) \leq_P \sigma(x_2)$ by the choice of x . The only other possibility is that $x_2 = x$, but then $\sigma(x_1) \leq_P \sigma(y)$ by the choice of x , and $\sigma(y) \leq \pi(\sigma(y)) = \sigma(x)$ by the expansiveness of π . Thus, (7-18) holds for x , which is a contradiction.

CASE 2. x is limit. By the choice of x ,

$$x_1 < x_2 < x \implies \sigma(x_1) \leq \sigma(x_2), \quad (7-19)$$

and so in order to get a contradiction, we only need show that

$$x_1 < x \implies \sigma(x_1) \leq_P \sigma(x).$$

This holds because (7-19) and the definition of σ also imply immediately that $\sigma(x) = \sup_P \{\sigma(y) \mid y < x\}$. $\dashv$ (Sublemma)

The result now follows directly, since the Sublemma implies in particular that the “otherwise” case in the definition of σ never comes up. $\dashv$

The transfinite orbit $\sigma : U \rightarrow P$ of a mapping $\pi : P \rightarrow P$ guaranteed by the Iteration Lemma is obviously an extension of the orbit $(n \mapsto x_n)$ which we defined in the proof of the Continuous Least Fixed Point Theorem 6.21, at least if the well ordered set U is longer than $\mathbb{N}$. It is one of the tools we will use in the proof of the Least Fixed Point Theorem, as follows.

7.26. Plan for a proof. Suppose that for the given inductive poset P , we can construct a well ordered set U such that *there exists no injection* $\sigma : U \rightarrowtail P$. In particular, the transfinite orbit $\sigma : U \rightarrow P$ of 7.25 cannot be an injection, and so there must exist $x < y$ such that $\sigma(x) = \sigma(y)$. The monotonicity of σ implies that

$$x \leq u \leq y \implies \sigma(x) = \sigma(u);$$

x has a successor since it is not maximum in U , $x < Sx \leq y$; and, hence,

$$\sigma(x) = \sigma(Sx) = \pi(\sigma(x)),$$

in other words, the point $\sigma(x)$ is a fixed point of π .

Thus, to prove that every expansive mapping $\pi : P \rightarrow P$ on an inductive poset has a fixed point, it is sufficient to show that *for each set P , there exists some well ordered set U which cannot be injected into P* . This is precisely Hartogs’ Theorem, for which we aim next. To show it, we must study in some detail the question of comparability of well ordered sets as to length.

The picture of the typical well ordered set in Figure 7.1 suggests that we should be able to compare any two of them, line them up side-by-side, the least element 0_U of one facing the least element 0_V of the other, the next $S_U(0_U)$ facing $S_V(0_V)$, the first limit point ω_U (if it exists) facing ω_V , etc. until we run out of elements in either U or V . The precise version of this fact is a generalization of the Uniqueness Theorem for the natural numbers 5.4.

7.27. Definition. An **initial similarity**

$$\pi : U \rightarrowtail \pi[U] \sqsubseteq V$$

from one well ordered set into another is a similarity of U with an initial segment of V . If such an initial similarity exists, we say that U is **less than or equal to V in length**, in symbols:

$$U \leq_o V \iff_{\text{df}} (\exists I \sqsubseteq V)[U =_o I]. \quad (7-20)$$

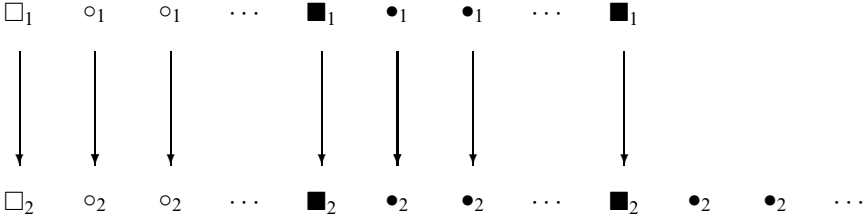


FIGURE 7.4. Portrait of an initial similarity.

We also write

$$U <_o V \iff_{\text{df}} U \leq_o V \ \& \ U \neq_o V. \quad (7-21)$$

By 7.9, every initial similarity $\pi : U \rightarrow V$ is either a similarity with V , or one with a proper initial segment of V , so that

$$U <_o V \iff (\exists x \in V)[U =_o \text{seg}_V(x)]. \quad (7-22)$$

7.28. Exercise. If $\pi : U \rightarrow V$ and $\rho : V \rightarrow W$ are initial similarities, then so is their composition $\rho\pi : U \rightarrow W$.

7.29. Proposition. For all well ordered sets U, V, W ,

$$\begin{aligned} &U \leq_o U, \\ &\text{if } [U \leq_o V \ \& \ V \leq_o W], \text{ then } U \leq_o W, \\ &\text{if } [U \leq_o V \ \& \ V \leq_o U], \text{ then } U =_o V. \end{aligned}$$

PROOF. Only the third of these assertions needs proof and it follows from 7.21. The composition $\rho\pi$ of the initial similarities $\pi : U \rightarrow V$, $\rho : V \rightarrow W$ witnessing the hypothesis is an initial similarity $\rho\pi : U \rightarrow W$, which if it were not onto, would witness that U is similar with one of its proper initial segments; so it is a bijection, and then π must also be a bijection. $\dashv$

7.30. Theorem. A function $\pi : U \rightarrow V$ is an initial similarity of a well ordered set into another if and only if it satisfies the identity

$$\pi(x) = \min_V \{y \in V \mid (\forall u <_U x)[\pi(u) <_V y]\} \quad (x \in U). \quad (7-23)$$

PROOF. If $\pi : U \rightarrow V$ is an initial similarity, then it is order-preserving and one-to-one, so it satisfies

$$(\forall u <_U x)[\pi(u) <_V \pi(x)], \quad (7-24)$$

and hence

$$z = \min_V \{y \in V \mid (\forall u <_U x)[\pi(u) <_V y]\} \leq_V \pi(x).$$

Since π is initial and $z \leq_V \pi(x)$, there exists some $u \in U$ such that $\pi(u) = z$. Assuming towards a contradiction that $z = \pi(u) <_V \pi(x)$, we infer that $u <_U x$ because π is an order-preserving injection; and hence $\pi(u) <_V z$ by the definition of z , which is absurd since $z = \pi(u)$.

Conversely, if $\pi : U \rightarrow V$ satisfies (7-23), then it is an order-preserving injection, since by (7-23), $u <_U x \implies \pi(u) <_V \pi(x)$. Suppose the image

$\pi[U]$ is not an initial segment of V and choose x least in U such that there exists some $y <_V \pi(x)$, $y \notin \pi[U]$. Now $\pi[\text{seg}_U(x)] \subseteq V$ by the choice of x ; it is a proper initial segment since it does not contain y ; so $\pi[\text{seg}_U(x)] = \text{seg}_V(z)$ for some $z \in V$, and (7-23) yields $\pi(x) = z$. Thus $y <_V z$ and $y \in \text{seg}_V(z) = \pi[\text{seg}_U(x)]$, which is absurd. $\dashv$

7.31. Theorem (Comparability of well ordered sets). *For any two well ordered sets U, V , either $U \leq_o V$ or $V \leq_o U$.*

PROOF. The result is trivial if $V = \emptyset$, so we may assume the minimum 0_V exists. By the Transfinite Recursion Theorem 7.24, there exists a function $\pi : U \rightarrow V$ which satisfies the identity

$$\pi(x) = \begin{cases} \min_V \{y \in V \mid (\forall u <_U x)[\pi(u) <_V y]\}, & \text{if } (\exists y \in V)(\forall u <_U x)[\pi(u) <_V y], \\ 0_V, & \text{otherwise.} \end{cases} \quad (7-25)$$

In pedantic detail, we are applying here Theorem 7.24 with the mapping $h : (U \rightarrow E) \times U \rightarrow E$, defined by

$$h(p, x) = \begin{cases} \min_V \{y \in V \mid (\forall u <_U x)[p(u) <_V y]\}, & \text{if } (\exists y \in V)(\forall u <_U x)[p(u) <_V y], \\ 0_V, & \text{otherwise.} \end{cases}$$

We now distinguish two possibilities.

CASE 1. *For every $x \neq 0_U$, $\pi(x) \neq 0_V$.* This means that the second case in (7-25) never applies, π satisfies the identity (7-23) and it must be an initial similarity by Theorem 7.30.

CASE 2. *For some $a \in U$, $a \neq 0_U$, we have $\pi(a) = 0_V$.* Let $a \neq 0_U$ be least in U and such that $\pi(a) = 0_V$, and consider the restriction

$$\rho = (\pi \upharpoonright \text{seg}_U(a)) : \text{seg}_U(a) \rightarrow V.$$

Now ρ satisfies (7-23), so by Theorem 7.30, it is an initial similarity of $\text{seg}_U(a)$ into V . In particular, the image $\rho[\text{seg}_U(a)] = \pi[\text{seg}_U(a)]$ is an initial segment of V ; if it were proper, then $\pi[\text{seg}_U(a)] = \text{seg}_V(z)$ for some $z \in V$ and (7-25) would yield $\pi(a) = z \neq 0_V$, contradicting the choice of a ; hence, $\pi[\text{seg}_U(a)] = V$. Thus, $V =_o \text{seg}_U(a)$, which gives us an initial similarity of V into U . $\dashv$

This fundamental theorem has a host of corollaries, some of which are worth listing immediately. The first one gives us an easier way to compare well ordered sets.

7.32. Corollary. *For all well ordered sets U, V ,*

$$U \leq_o V \iff (\exists \pi : U \rightarrow V)[\pi \text{ is order-preserving}].$$

PROOF. Suppose $\pi : U \rightarrow V$ is order-preserving but $U \not\leq_o V$, so that $V <_o U$. It follows that $V =_o \text{seg}_U(x)$ for some x by (7-22), and composing the order-preserving injections we get an injection $\rho : U \rightarrow \text{seg}_U(x)$ which is still order-preserving and violates 7.20. $\dashv$

7.33. Corollary (Wellfoundedness of $\leq_o$). *Every non-empty class $\mathcal{E}$ of well ordered sets has a $\leq_o$ -least member, i.e., for some $U_0 \in \mathcal{E}$ and all $U \in \mathcal{E}$, $U_0 \leq_o U$.*

PROOF. The hypothesis gives us some $W \in \mathcal{E}$, and if W is $\leq_o$ -least in $\mathcal{E}$, there is nothing to prove. If not, then 7.31 implies that there exists well ordered sets in $\mathcal{E}$ which are similar with proper initial segments of W , so the set

$$J =_{\text{df}} \{x \in W \mid (\exists U \in \mathcal{E})[U =_o \mathbf{seg}_W(x)]\} \quad (7-26)$$

is non-empty and it has a $\leq_W$ -least element x . By the definition of J , there exist some $U_0 \in \mathcal{E}$ such that $U_0 =_o \mathbf{seg}_W(x)$ and we claim that this U_0 is $\leq_o$ -least in $\mathcal{E}$. To prove it, assume towards a contradiction that for some $U \in \mathcal{E}$, $U_0 \not\leq_o U$; hence $U <_o U_0 =_o \mathbf{seg}_W(x)$; hence $U =_o \mathbf{seg}_W(y)$ for some $y <_W x$, contradicting the choice of x . $\dashv$

Most often this is applied when $\mathcal{E}$ is actually a set, a family of well ordered sets, but occasionally it is convenient to cite it more generally for classes. For example, there exists a $\leq_o$ -least well ordered set which has a limit point—namely $\text{Succ}(\mathbb{N})$.

After all this work, still we have not constructed any uncountable well ordered sets and it might appear that all our results apply only to peculiar, long reshufflings of $\mathbb{N}$. Next comes the second basic theorem of this chapter which rectifies the situation.

7.34. Hartogs' Theorem. *There is a definite operation $\chi(A)$ which associates with each set A , a well ordered set*

$$\chi(A) = (h(A), \leq_{\chi(A)}),$$

such that $h(A) \not\leq_c A$, i.e., there exists no injection $\pi : h(A) \hookrightarrow A$. Moreover, $\chi(A)$ is $\leq_o$ -minimal with this property, i.e., for every well ordered set W ,

$$\text{if } W \not\leq_c A, \text{ then } \chi(A) \leq_o W. \quad (7-27)$$

PROOF. First set

$$\begin{aligned} \text{WO}(A) =_{\text{df}} \{U \mid U = (\text{Field}(U), \leq_U) \text{ is a well ordered set} \\ \text{with } \text{Field}(U) \subseteq A\}, \end{aligned} \quad (7-28)$$

and let $\sim_A$ be the restriction of the definite condition $=_o$ to $\text{WO}(A)$,

$$U \sim_A V \iff_{\text{df}} U, V \in \text{WO}(A) \text{ \& } U =_o V.$$

Clearly $\sim_A$ is an equivalence relation on $\text{WO}(A)$, and we set

$$h(A) =_{\text{df}} [\text{WO}(A)/\sim_A] \subseteq \mathcal{P}(\text{WO}(A)). \quad (7-29)$$

We order the equivalence classes in $h(A)$ by their “representatives”,

$$[U/\sim_A] \leq_{\chi(A)} [V/\sim_A] \iff_{\text{df}} U \leq_o V; \quad (7-30)$$

this makes sense because if

$$[U/\sim_A] = [U'/\sim_A], [V/\sim_A] = [V'/\sim_A], \text{ and } U \leq_o V,$$

then $U' =_o U \leq_o V =_o V'$. The fact that $\leq_{\chi(A)}$ is a wellordering of $h(A)$ follows easily from the general properties of $\leq_o$, 7.31 and 7.33. Taking the negation of both sides of (7-30) we infer its strict version,

$$V <_o U \iff [V/\sim_A] <_{\chi(A)} [U/\sim_A] \quad (U, V \in \mathbf{WO}(A)). \quad (7-31)$$

The basic properties of the Hartogs operation are embodied in the following

Lemma. *For every $\alpha = [U/\sim_A] \in h(A)$,*

$$\mathbf{seg}_{\chi(A)}(\alpha) = \{[\mathbf{seg}_U(x)/\sim_A] \mid x \in U\} =_o U.$$

In particular, every proper initial segment of $\chi(A)$ is similar with some well ordered set $U \in \mathbf{WO}(A)$, and every $U \in \mathbf{WO}(A)$ is similar with a proper, initial segment of $\chi(A)$.

Proof. We verify first the identity

$$\mathbf{seg}_{\chi(A)}(\alpha) = \{[\mathbf{seg}_U(x)/\sim_A] \mid x \in U\}.$$

If $\beta = [V/\sim_A] <_{\chi(A)} \alpha$, then $V <_o U$ from (7-31), and hence $V =_o \mathbf{seg}_U(x)$ for some $x \in U$, so that $\beta = [\mathbf{seg}_U(x)/\sim_A]$. Conversely, for each $x \in U$, $\mathbf{seg}_U(x) <_o U$, hence, $[\mathbf{seg}_U(x)/\sim_A] <_{\chi(A)} [U/\sim_A] = \alpha$, again by (7-31).

To show the similarity

$$U =_o \mathbf{seg}_{\chi(A)}(\alpha) = \{[\mathbf{seg}_U(x)/\sim_A] \mid x \in U\},$$

define $\rho : U \rightarrow h(A)$ by

$$\rho(x) = [\mathbf{seg}_U(x)/\sim_A], \quad (x \in U);$$

now ρ is a similarity of U with the image $\rho[U]$, because

$$\begin{aligned} x <_U y &\iff \mathbf{seg}_U(x) \subsetneq \mathbf{seg}_U(y) \\ &\iff \mathbf{seg}_U(x) <_o \mathbf{seg}_U(y) \\ &\iff [\mathbf{seg}_U(x)/\sim_A] <_{\chi(A)} [\mathbf{seg}_U(y)/\sim_A]. \quad \neg (\text{Lemma}) \end{aligned}$$

Suppose now, towards a contradiction, that there exists an injection

$$\pi : h(A) \hookrightarrow A,$$

and let $B = \pi[h(A)] \subseteq A$ be its image. The injection π copies the wellordering of $h(A)$ to a wellordering of B ,

$$x \leq_B y \iff_{\text{df}} \pi^{-1}(x) \leq_{\chi(A)} \pi^{-1}(y) \quad (x, y \in B),$$

so that $U = (B, \leq_B)$ is a well ordered subset of A , and by its definition,

$$U =_o \chi(A). \quad (7-32)$$

But U is similar with a proper initial segment of $\chi(A)$ by the Lemma, and hence $U <_o \chi(A)$, which contradicts $U =_o \chi(A)$.

To show the minimality of $\chi(A)$, notice that for any well ordered set W , if $W <_o \chi(A)$, then $W =_o \mathbf{seg}_{\chi(A)}(\alpha)$ for some $\alpha = [U/\sim_A]$, so that $W =_o U$ by the Lemma. Thus,

$$\text{if } W <_o \chi(A), \text{ then } W \leq_c A, \quad (7-33)$$

since (the field of) U is a subset of A and similarities are injections. Taking the negation of both sides,

$$\text{if } W \not\leq_c A, \text{ then } \neg[W <_o \chi(A)], \text{ hence } \chi(A) \leq_o W. \quad \neg$$

Of course, we would like to prove that $A <_c \chi(A)$ instead of the timid $\chi(A) \not\leq_c A$, and this is certainly true, but its proof depends on the Axiom of Choice. Wait for a bit until we finally bring the Deus ex Machina onto the stage.

The annoying details of this proof are forced on us by the fact that the restriction $\lesssim_A$ of the definite condition $\leq_o$ to $\text{WO}(A)$ is not a wellordering, for the trivial reason that it is not antisymmetric: there may exist distinct, similar $U, V \in \text{WO}(A)$, in fact they always do, if A has more than one element. This is why we were forced to take $h(A)$ as a set of equivalence classes rather than simply set $h(A) = \text{WO}(A)$. Technically, $\lesssim_A$ is a *prewellordering* (ugh!) and it is worth recasting the argument in a different form, after introducing this notion. See Problems **x7.17** – **x7.20**.

The Hartogs operation can be used to construct general *infimum* and *supremum* operation for families of well ordered sets (Problems **x7.26** and **x7.27**), and it has many other interesting properties. We use it next to extend the Continuous Least Fixed Point Theorem **6.21** to discontinuous mappings. Let us first put down, for the record, the Fixed Point Theorem for expansive mappings, which we have already discussed.

7.35. Fixed Point Theorem (Zermelo).¹⁸ *Every expansive mapping $\pi : P \rightarrow P$ on an inductive poset has at least one fixed point, i.e., some $x^* \in P$ satisfies the equation*

$$x^* = \pi(x^*).$$

PROOF. The argument given in **7.26** needs only some well ordered set U which cannot be injected into P , and $U = \chi(P)$ does it. $\neg$

7.36. Least Fixed Point Theorem. *Every monotone mapping $\pi : P \rightarrow P$ on an inductive poset has a (unique) least fixed point, i.e., for some $x^* \in P$,*

$$\pi(x^*) = x^*,$$

$$(\forall y \in P)[\pi(y) = y \implies x^* \leq y].$$

In fact, x^ satisfies the following, strong minimality property:*

$$(\forall y \in P)[\pi(y) \leq y \implies x^* \leq y]. \quad (7-34)$$

PROOF. A careful examination of the proof of the Iteration Lemma **7.25** and the proof of the Fixed Point Theorem **7.26** reveals that exactly the same construction of the fixed point for an expansive mapping works and yields

¹⁸Zermelo did not formulate the Fixed Point Theorem in this generality, which is why it and many of its Corollaries have been attributed at various times to later mathematicians. But the famous “first proof” of the *Wellordering Theorem* which Zermelo gave in 1904 proves exactly this result, trivially restricted to the special case which interested him.

the least fixed point of a monotone mapping. However, it is not necessary to do this, as the Least Fixed Point Theorem is an easy consequence of the Fixed Point Theorem. The basic idea is to observe that the given monotone mapping π is necessarily expansive on some inductive *sub-poset* of P .

Let

$$Q = \{x \in P \mid x \leq \pi(x) \ \& \ (\forall y)[\pi(y) \leq y \implies x \leq y]\}$$

and observe first that the restriction

$$\leq_Q = \{(x, y) \mid x, y \in Q \ \& \ x \leq_P y\}$$

of $\leq_P$ to Q is also a partial ordering—this is automatically true for the restriction of $\leq_P$ to any subset of P . (We skip the subscripts P and Q for the remaining of the argument.) In addition, $\pi[Q] \subseteq Q$, because

$$x \leq \pi(x) \implies \pi(x) \leq \pi(\pi(x)),$$

and for every y ,

$$\pi(y) \leq y \ \& \ x \leq y \implies \pi(x) \leq \pi(y) \leq y,$$

by the monotonicity of π . It follows that the restriction

$$\pi_Q = \{(x, \pi(x)) \mid x \in Q\}$$

of π to Q is a mapping on Q , it continues to be monotone (of course) and it is also expansive, because of the definition of Q . To apply the Fixed Point Theorem 7.35 to Q and π_Q we need the following.

Lemma. *The poset Q is inductive.*

Proof. It is enough to show that for every chain $S \subseteq Q$, the least upper bound $M = \sup S$ (which exists in P because P is inductive) is a member of Q , i.e., (1) $M \leq \pi(M)$, and (2) for every y , $\pi(y) \leq y \implies M \leq y$. For (1) we compute:

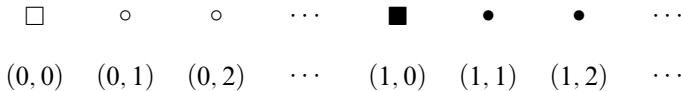
$$\begin{aligned} x \in S &\implies x \leq M, && \text{because } M \text{ is an upper bound of } S, \\ \implies \pi(x) &\leq \pi(M), && \text{because } \pi \text{ is monotone,} \\ \implies x &\leq \pi(x) \leq \pi(M), && \text{because } x \in S \subseteq Q, \end{aligned}$$

and therefore $\pi(M)$ is an upper bound of S and we have $M = \sup S \leq \pi(M)$. (2) follows from the observation that every y such that $\pi(y) \leq y$ is an upper bound of Q (from Q 's definition), and therefore an upper bound of the smaller set $S \subseteq Q$, so that $M = \sup S \leq y$. $\dashv$ (Lemma)

By the Fixed Point Theorem 7.35 now, there exists some $x^* \in Q$, such that $\pi(x^*) = x^*$ and (7-34) holds simply because $x^* \in Q$.

That there is at most one least fixed point is obvious: if y^* is also a least fixed point, then $x^* \leq y^*$ and $y^* \leq x^*$, so $x^* = y^*$. $\dashv$

The full Least Fixed Point Theorem frees us from the necessity to check continuity in the applications of least fixed points to computer science, the fixpoint theory of programs. This is nice, but not very important, since (as we observed in Chapter 6) the mappings which come up in algorithmic applications are typically continuous “by inspection”. However, the theorem has

FIGURE 7.5. The sum $\mathbb{N} +_o \mathbb{N}$.

more significant, deeper applications to the general theory of sets, particularly in the study of *definability* in set theory as well as the construction of examples and counterexamples with specified properties. We will encounter several of these in the chapters which follow.

Problems for Chapter 7

x7.1. Every linear ordering of a finite set is a wellordering. (See the related Problem **x6.7**.)

7.37. The **sum** $P +_o Q$ of two posets P and Q is obtained by placing disjoint copies of P and Q side-by-side, every point of P preceding every point of Q . Formally, we set $P +_o Q = R$, where

$$\text{Field}(R) =_{\text{df}} (\{0\} \times \text{Field}(P)) \cup (\{1\} \times \text{Field}(Q)), \quad (7-35)$$

and for $(i, x), (j, y) \in \text{Field}(R)$,

$$(i, x) \leq_R (j, y) \iff_{\text{df}} i < j \vee [i = j = 0 \ \& \ x \leq_U y] \vee [i = j = 1 \ \& \ x \leq_V y]. \quad (7-36)$$

The idea is that P is similar with the set $\{0\} \times \text{Field}(P)$ partially ordered by its second elements, by the obvious similarity $(x \mapsto (0, x))$, and (again) $Q =_o \{1\} \times \text{Field}(Q)$.

x7.2. If $P =_o P'$ and $Q =_o Q'$, then $P +_o Q =_o P' +_o Q'$.

x7.3. For all posets P , $\text{Succ}(P) =_o P +_o [0, 1)$.

x7.4. For all posets P, Q, R ,

$$P +_o (Q +_o R) =_o (P +_o Q) +_o R.$$

x7.5. If U and V are well ordered sets, then so is their sum $U +_o V$.

x7.6. Prove that $[0, 1) +_o \mathbb{N} =_o \mathbb{N} \not=_o \mathbb{N} +_o [0, 1)$, so that the addition operation on well ordered sets is not commutative.

7.38. The **product** $P \cdot_o Q$ of two posets is obtained by replacing each point of Q by a copy of P . Formally, we let $P \cdot_o Q = R$, where

$$\text{Field}(R) = \text{Field}(P) \times \text{Field}(Q), \quad (7-37)$$

and $\leq_R$ is the *inverse lexicographic* ordering of pairs i.e., we compare the second members first: for $(x_1, y_1), (x_2, y_2) \in \text{Field}(R)$,

$$(x_1, y_1) \leq_R (x_2, y_2) \iff_{\text{df}} y_1 <_Q y_2 \vee [y_1 = y_2 \ \& \ x_1 \leq_P x_2]. \quad (7-38)$$

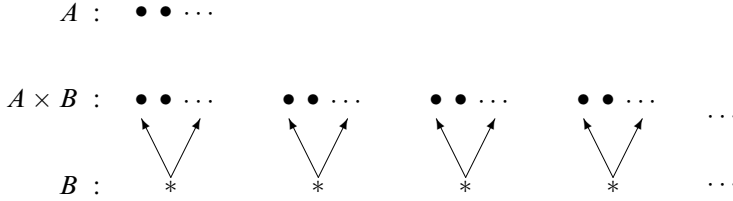


FIGURE 7.6. The product of two well ordered sets.

There is no special reason for ordering pairs by looking at their second members first, it is just that Cantor chose to do it this way and it has stuck.

x7.7. If $P =_o P'$ and $Q =_o Q'$, then $P \cdot_o Q =_o P' \cdot_o Q'$.

x7.8. Prove that $P \cdot_o [0, 2) =_o P +_o P$, but $[0, 2) \cdot_o \mathbb{N} =_o \mathbb{N} \neq_o \mathbb{N} \cdot_o [0, 2)$, so that multiplication of well ordered sets is not commutative.

x7.9. For all posets P, Q, R ,

$$P \cdot_o (Q \cdot_o R) =_o (P \cdot_o Q) \cdot_o R.$$

x7.10. The product of two well ordered sets is well ordered.

x7.11. For each well ordered set U , there exists exactly one function

$$\text{Parity} : U \rightarrow \mathbb{N},$$

such that $\text{Parity}(y) = 0$ if $y = 0$ or y is a limit point, and at successor points,

$$\text{Parity}(S(x)) = 1 - \text{Parity}(x).$$

x7.12. Every point y in a well ordered set U can be expressed uniquely in the form

$$y = S^n(x), \tag{7-39}$$

where (1) x is either the minimum 0 or a limit point, (2) n is a natural number, and (3) the function $(i, x) \mapsto S^i(x)$ is defined by the recursion

$$S^0(x) = x, \quad S^{i+1}(x) = S(S^i(x)).$$

x7.13. For any two well ordered sets U, V , there exists at most one initial similarity $\pi : U \rightarrow \pi[U] \subseteq V$.

x7.14. For all well ordered sets U, V, W ,

$$U <_o V \ \& \ V \leq_o W \implies U <_o W,$$

$$U \leq_o V \ \& \ V <_o W \implies U <_o W.$$

x7.15. If κ and λ are well orderable cardinal numbers, then either $\kappa \leq_c \lambda$ or $\lambda \leq_c \kappa$.

***x7.16.** If κ is a well orderable, infinite cardinal number, then $\kappa + 1 =_c \kappa$.

	(0, 0)	(1, 0)	(2, 0)	...	(0, 1)	(1, 1)	(2, 1)	...
$N \cdot_o [0, 2) :$	□	○	○	...	■	●	●	...
$[0, 2) \cdot_o N :$	□	○	○	○	○	...		
	(0, 0)	(1, 0)	(0, 1)	(1, 1)	(0, 2)	...		

FIGURE 7.7. Multiplication of well ordered sets is not commutative.

7.39. Definition. A **prewellordering** on a set A is any relation $\lesssim \subseteq A \times A$ which is reflexive, transitive, connected (total) and grounded. “Connected” means that any two points in A are comparable,

$$(\forall x, y \in A)[x \lesssim y \vee y \lesssim x],$$

and “grounded” means that every non-empty $X \subseteq A$ has a $\lesssim$ -least member,

$$(\forall X \subseteq A, X \neq \emptyset)(\exists x \in X)(\forall y \in X)[x \lesssim y].$$

A prewellordering would be a wellordering, if only it were antisymmetric.

x7.17. For each set A , consider the set

$$B = \{X \subseteq A \mid X \text{ is finite}\}$$

of all finite subsets of A , and set on B

$$X \lesssim_B Y \iff_{\text{df}} X \leq_c Y.$$

Prove that $\lesssim_B$ is a prewellordering.

x7.18. A relation $\lesssim \subseteq A \times A$ is a prewellordering if and only if there exists a well ordered set $U = (\text{Field}(U), \leq_U)$ and a surjection $\pi : A \twoheadrightarrow \text{Field}(U)$ such that

$$x \lesssim y \iff \pi(x) \leq_U \pi(y) \quad (x, y \in A).$$

x7.19. For each set A , the relation

$$U \lesssim_A V \iff_{\text{df}} U, V \in \mathbf{WO}(A) \ \& \ U \leq_o V$$

is a prewellordering of $\mathbf{WO}(A)$.

x7.20. Rework the proof of Hartogs’ Theorem by applying the preceding two problems.

x7.21. For every set A , there exists a well ordered set V such that there exists no surjection $\pi : A \twoheadrightarrow V$.

x7.22. Prove that if $A \leq_c B$, then $\chi(A) \leq_o \chi(B)$.

x7.23. Prove that $\chi([0, n)) =_o [0, n + 1)$.

x7.24. If W is a well ordered set and $W \leq_c A$, then $W <_o \chi(A)$.

x7.25. For each set A and each well ordered set U ,

$$U <_o \chi(A) \iff \text{Field}(U) \leq_c A.$$

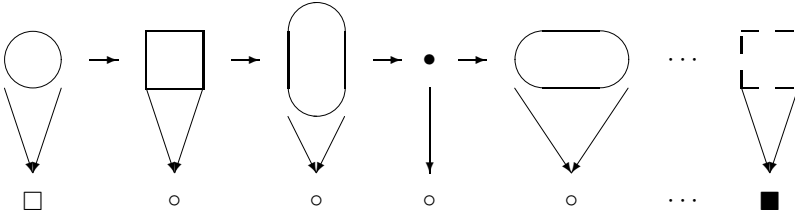


FIGURE 7.8. Portrait of a prewellordering.

The operation $\chi(A)$ is definite, as we gave an explicit definition of the field $h(A)$ and the wellordering $\leq_{\chi(A)}$ of $\chi(A)$ from A . We can use it to define some related, “infinitary” operations on families of well ordered sets.

***x7.26.** Define a definite operation $\inf(\mathcal{E})$, such that for every non-empty family $\mathcal{E}$ of well ordered sets, $\inf(\mathcal{E})$ has the following properties:

- (1) $\inf(\mathcal{E})$ is a well ordered set.
- (2) For some $U \in \mathcal{E}$, $\inf(\mathcal{E}) =_o U$.
- (3) For every $U \in \mathcal{E}$, $\inf(\mathcal{E}) \leq_o U$.

HINT. Look for $\inf(\mathcal{E})$ in the initial segments of $\chi(\bigcup \mathcal{E})$.

***x7.27.** Define a definite operation $\sup(\mathcal{E})$, such that for every non-empty family $\mathcal{E}$ of well ordered sets, $\sup(\mathcal{E})$ has the following properties:

- (1) $\sup(\mathcal{E})$ is a well ordered set.
- (2) If $U \in \mathcal{E}$, then $U \leq_o \sup(\mathcal{E})$.
- (3) If W is a well ordered set and for each $U \in \mathcal{E}$, $U \leq_o W$, then $\sup(\mathcal{E}) \leq_o W$.

***x7.28.** Let $\leq$ be a linear ordering of a set A and define on the poset $\mathcal{P}(A)$ the mapping

$$\pi(X) =_{\text{df}} \{y \in A \mid (\forall x < y)[x \in X]\}.$$

Verify that $\pi : \mathcal{P}(A) \rightarrow \mathcal{P}(A)$ is monotone and give an example where it is not countably continuous. Prove that if A_w is the least fixed point of π , then

$$x \in A_w \iff \{(s, t) \in A \times A \mid s \leq t < x\} \text{ is a wellordering.}$$

There are situations where it is easier to use *the proof* of the Fixed Point Theorem 7.35 rather than its statement.

***x7.29. Detailed Fixed Point Theorem.** For each expansive or monotone mapping $\pi : P \rightarrow P$ on an inductive poset P , there exists a subset $D \subseteq P$ with the following properties:

1. D is a well ordered chain in P .
2. Every member of D is determined from its predecessors by the formula

$$x = \pi(\sup \{y \in D \mid y < x\}).$$

3. No point in D is a fixed point of π .

4. The point $\pi(\sup D)$ is a fixed point of π ,

$$\pi(\pi(\sup D)) = \pi(\sup D).$$

5. If π is monotone, then $\pi(\sup D)$ is the least fixed point of π .

Prove also that these conditions determine D uniquely.

***x7.30.** Suppose P and Q are inductive posets and $\pi : P \times Q \rightarrow P$ is a monotone mapping on the product and define the mapping $\rho : Q \rightarrow P$ by appealing to Problem **x6.4** and the Least Fixed Point Theorem **7.36**,

$$\begin{aligned} \rho(y) &= (\mu x \in P)[\pi(x, y) = x] \\ &= \text{the least fixed point of } \pi(x, y) = x. \end{aligned} \tag{7-40}$$

Prove that ρ is a monotone mapping, and if π is countably continuous, then so is ρ .

***x7.31. Bekič-Scott Rule.** Suppose P_1, P_2 are inductive posets, and

$$\pi_1 : P_1 \times P_2 \rightarrow P_1, \quad \pi_2 : P_1 \times P_2 \rightarrow P_2$$

are monotone mappings. Using the μ -notation for least fixed points of (7-40), let

$$\rho(x_2) = (\mu x_1 \in P_1)[\pi_1(x_1, x_2) = x_1],$$

let

$$\overline{x}_2 = (\mu x_2 \in P_2)[\pi_2(\rho(x_2), x_2) = x_2]$$

be the least fixed point of the mapping $x_2 \mapsto \pi_2(\rho(x_2), x_2)$ (which is monotone by **x7.30**) and finally let

$$(x_1^*, x_2^*) = (\mu(x_1, x_2) \in P_1 \times P_2)[(\pi_1(x_1, x_2), \pi_2(x_1, x_2)) = (x_1, x_2)]$$

be the least fixed point in the product poset. Prove that

$$x_2^* = \overline{x}_2.$$

The problem insures that we can compute simultaneous least fixed points by iterating the least fixed point operation $(\mu x \in P)$ on one inductive poset at a time.